Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design

Generation Category (1): Design

Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	1	0	2	3

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	0	0	0	0	2	0	2

Method of Solution:	Newton-Raphson Method
Maximum No. of Iteration:	3
Precision of Solution:	0.0001000
System Frequency:	50.00 Hz
Unit System:	Metric
Project Filename:	NR_Method
Output Filename:	C:\Users\Eng Abdalsamie\Desktop\ETAP - Projects\Load Flow Analysis\Newton Raphson Method\ETAP 19 Project\NR_Method.lfr

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	1.050	1	100.0	0.0								
Bus2	1.000	1	100.0	0.0	0.257	0.110						
Bus3	1.000	1	100.0	0.0	0.139	0.044						
Total Number of Buses: 3					0.395	0.154	0.000	0.000	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	1.050	Swing	1	100.0	0.0					
						0.000	0.000			

Impedance Input Data

Impedance	Positive Sequence Impedance			
ID	R	X	Y	Unit
Z1	2	4	0	% in 1.000 kV base and 1.0 MVA base
Z3	1	3	0	% in 1.000 kV base and 1.0 MVA base

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
Z1	Impedance	Bus2	Bus1	181.41	362.81	405.64	
Z3	Impedance	Bus3	Bus1	90.70	272.11	286.83	

LOAD_FLOW_REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
* Bus1	1.050	100.000	0.0	0.397	0.158	0.000	0.000	Bus2	0.258	0.113	154.9	91.6	
								Bus3	0.139	0.045	80.2	95.2	
Bus2	1.000	104.081	-0.4	0.000	0.000	0.257	0.110	Bus1	-0.257	-0.110	154.9	91.9	
Bus3	1.000	104.740	-0.2	0.000	0.000	0.139	0.044	Bus1	-0.139	-0.044	80.2	95.3	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

\# Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					Percent
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Loading
Bus1	1.050										0.427	92.9	234.8	
Bus2	1.000		0.257	0.110							0.279	91.9	154.9	
Bus3	1.000		0.139	0.044							0.146	95.3	80.2	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).
Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Z1	0.258	0.113	-0.257	-0.110	1.4	2.9	100.0	104.1	0.92
Z3	0.139	0.045	-0.139	-0.044	0.2	0.6	100.0	104.7	0.26
					1.6	3.5			

* This Transmission Line includes Series Capacitor.

Alert Summary Report

	% Alert Settings	
	Critical	Marginal
Loading		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
Bus Voltage		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
Generator Excitation		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus2	Bus	Over Voltage	1.000	kV	1.041	104.1	3-Phase
Bus3	Bus	Over Voltage	1.000	kV	1.047	104.7	3-Phase

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	0.397	0.158	0.427	92.93 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	0.397	0.158	0.427	92.93 Lagging
Total Motor Load:	0.395	0.154	0.424	93.16 Lagging
Total Static Load:	0.000	0.000	0.000	
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.002	0.003		
System Mismatch:	0.000	0.000		

Number of Iterations: 1